Ana I. Pereira · Florbela P. Fernandes ·
João P. Coelho · João P. Teixeira ·
Maria F. Pacheco · Paulo Alves ·
Rui P. Lopes (Eds.)

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Editors
Ana I. Pereira
Instituto Politécnico de Bragança
Bragança, Portugal

João P. Coelho
Instituto Politécnico de Bragança
Bragança, Portugal

Maria F. Pacheco
Instituto Politécnico de Bragança
Bragança, Portugal

Rui P. Lopes
Instituto Politécnico de Bragança
Bragança, Portugal

Florbela P. Fernandes
Instituto Politécnico de Bragança
Bragança, Portugal

João P. Teixeira
Instituto Politécnico de Bragança
Bragança, Portugal

Paulo Alves
Instituto Politécnico de Bragança
Bragança, Portugal

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Using Natural Language Processing for Phishing Detection

Richard Adolph Aires Jonker(✉), Roshan Poudel, Tiago Pedrosa, and Rui Pedro Lopes

Research Centre in Digitalization and Intelligent Robotics (CeDRI), Instituto Politécnico de Bragança, Bragança, Portugal
{richard,roshan,pedrosa,rlopes}@ipb.pt

Abstract. We live in a world where computers are constantly changing the way we do things. People spend many hours on their phones or computers, whether it be for work or leisure purposes. The danger is that these unsuspecting users can be targeted for attacks at any time and can fall victim to many types of scams or phishing attacks. These attacks can be harmful to the user by getting valuable credentials, money or even installing malicious software on their devices, all while the user is unaware of what has just happened. In a business environment these can lead to mass data breeches which could end up costing a company millions of euros. Many users are not trained to recognize phishing texts, so an alternative solution is needed to help prevent users from falling into these traps. In this paper we will be investigating Natural Language Processing (NLP), a subsection of Machine Learning (ML) to try generate solutions to the problem of phishing. We will investigating different NLP solutions: Word2Vec, Doc2Vec and BERT, and different ML solutions: RNN, LSTM, CNN and TD-IDF. All of these different approaches provide good classification results ranging from f1-scores of 90.03–98.94.

Keywords: Cybersecurity · Machine Learning · Artificial intelligence · Phishing · Natural Language Processing

1 Introduction

Since the creation of the commercial computer, the convergence to the digital age began. People have tried to move any task they could find to a digital media, this varies from business tasks, shopping and even social interactions. However, with this trend into the digital era, it also brought crime and criminal activities. As people have been moving towards computers, others have simultaneously been trying to exploit anything they can get their hands on. This led to one of the most dynamic fields in the computer industry, Cybersecurity. With the development of new software and algorithms, there will be people who try to exploit the system. This is just human nature. In the last few years, people in the cybersecurity field have tried to revolutionize the field using the help of ML

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and artificial intelligence [1]. This has greatly helped eliminate large amounts of similar threats, as ML excels in identifying complex patterns and processing efficiently large amounts of data.

ML has been tackling some of the biggest cyberthreats. ML is also one of the newest developments in the informatics field, and is slowly changing every other field. With the introduction of ML in the cybersecurity, many tasks have been automated, as people are trying to develop automatic solutions for many problems. Furthermore, some of these solutions are more cost effective than current market solutions. Even with this trend of the automation of the cybersecurity field, there are still many threats that remain. This paper approaches one problem that does not target the system, but rather the users, namely Phishing.

Phishing is a large issue in the cybersecurity field, as it cannot be directly countered. The burden to detect phishing falls on an unsuspecting user, which is why the success of these attacks are so high [2]. Generally the only way to counteract these types of attacks is user training, where the user can hopefully learn to differentiate these types of attacks. However, this method has been shown to be ineffective, as users need to be reminded about these threats every few months [3].

In the past few years, however, there have been a few breakthroughs in the NLP area of ML. This field revolves around the analysis and processing of the written word, which can help translate a human language into something a computer can understand. We have realised that it could be possible to use NLP to try and understand what classifies a phishing text from non phishing text which could result in a major contribution in the cybersecurity field, as we could use NLP to move the burden of phishing detection away from the user and place it on the machine instead.

1.1 Objectives

This study evaluates the effectiveness of using ML algorithms, more specifically, NLP algorithms, on phishing texts, to try and create a usable model for classifying phishing texts from non-phishing texts. We will be exclusively doing this for the English language, however the same approach could be used for additional languages. Using this study we can determine which NLP algorithm is best suited for phishing attacks. This information could be very useful for implementing solutions to this problem. To achieve our goal we have set the following objectives:

- Research the relevant fields of ML, NLP and phishing.
- Evaluate which NLP models will be relevant for our research area.
- Gather and process phishing texts for use in the models.
- Create models that will be suitable for phishing text classification.
- Evaluate the effectiveness of each of these models and determine which is the best suited for our use case.

1.2 Context and Technologies

The global internet population is increasing day by day and the modern world is becoming more and more dependent on the internet for everyday living. In January 2020, the number of people around the world using the internet has grown to 4.54 billion, an increase of 7% (298 million new users) compared to January 2019 [4]. With the expansion of cyberspace, the risk of being a victim of cyberattacks is also increasing. Cybersecurity has become the most important element in the cyberspace to battle against all threats, attacks, and frauds.

Cybersecurity refers to techniques, technologies and practices of defending devices, networks, programs and data from malicious attacks and unauthorized access [5]. Usually cyberattackers aim at getting sensitive information or money from users, or interrupting normal business processes.

In 2020, a Twitter breach targeted 130 accounts, including those of past presidents and Elon Musk, resulted in attackers swindling $121,000 in Bitcoin through nearly 300 transactions [6]. Attacks such as these are being more and more common day by day. These days, implementing an effective cybersecurity measure is challenging because of the sheer number of devices and attackers being more innovative.

Machine Learning is a branch of Artificial Intelligence that gives a machine the ability to learn and improve itself from its own experience without any explicit programming [7]. ML starts with machine finding patterns in the data or observations and using those to improve decisions in the future. Lately every industry is using ML to get data driven results which provides the industry with better results. New applications for ML are being found everyday due to its adaptability, scalability and potential to rapidly find solutions to new and unknown problems.

1.3 Phishing

Phishing is generically defined as using a medium to impersonate a person or organization, to deceive a user into giving confidential information away. These types of attacks can be spread on emails, SMS, social network messages or telephone calls. They try to attack the user rather than the system, as systems are generally secure, whereas the average user may not know what a phishing attack is, let alone how to identify one. It remains one of the biggest security threats in the cybersecurity field.

Phishing attacks try to get confidential user data such as usernames and passwords or banking details via either direct response to the attack, malicious links or malware executables. In bigger companies they can even be used to get money transferred into private accounts, by spoofing people who were supposed to get paid. The malware executables could be a form of ransomware, spyware, Trojan or even just a backdoor access into a system. Phishing attacks are only limited by the creativity of the attacker, and the user they are attacking. Personal attacks would be different from company attacks, as in the latter the goal generally would be to infiltrate the company, as opposed to just one user, netting in more destruction or higher financial gain.

2 Natural Language Processing

In a broad sense, NLP is the processing and manipulation of a language using software [8]. This allows computers to understand and interpret a language so that more precise processing can be done on the semantics of the language. We have decided to use NLP to try and discover what characterizes phishing texts, and how to differentiate them from normal texts. Hopefully a device can warn a user of phishing scams and help take the burden off the user to detect these scams, allowing the user to have a more free browsing experience. This approach could be advantageous in the cybersecurity field, allowing companies to save millions of dollars on training employees and preventing data breeches from phishing. Unfortunately, a machine is only as good as the data you provide it, so it will be very difficult to produce something that can correctly identify every single phishing text, and it will be more likely to get a few incorrectly identified results, such as false positives and negatives.

A lot of NLP algorithms are based on machine learning or statistical algorithms. In our project we aim to explore as many algorithms as we can. We have listed a few which we plan on using.

2.1 Word2Vec

Word2Vec tries to create associations between words based off a large text source which the model is trained off of [9]. The associations are based off the vector spaces of words which the model produces. These are known as word embeddings. In this model each word is represented as a dimension and the similarity between a word is based off the cosine distance between the 2 word vectors. This allows the model to find words used in similar context based off the vector values, and allows the computer to do arithmetic with these vectors to allow it to process and interpret the words and sentences. This allows us to do operations such as:

- Lisbon - Portugal + France = Rome
- King - Man + Women = Queen

2.2 Doc2Vec

Doc2Vec produces a numeric representation of a document [10]. It is based off Word2Vec, being similar to CBOW, however, an extra document specific vector is added which is document unique. When the document is trained, this additional document field is trained as well, and at the end this holds a value representing the document. Similar to a word vector, this represents a value unique to the document, based off its words.

With Doc2vec we can represent complete text as a vector and plot them in a vector space, which is a way of taking an unsupervised learning approach to the text classification. When new text document is fed into the model for classification it can provide output based on the distance between new document vectors and clusters of already plotted vectors.

2.3 Recurrent Neural Network (RNN)

A RNN is a class of neural network which takes the previous outputs as an input for the next generation whilst still having hidden states. This allows future generations to be influenced by what has already been discovered, however this is only limited to a few generations, due to a lack of resources and the long-term dependency problem. RNN needs to be fed sequential data, so the process cannot be parallelized, and thus it is much slower than other approaches. All RNN have a chain of repeating modules [11] (See Fig. 1 for graphic representation).

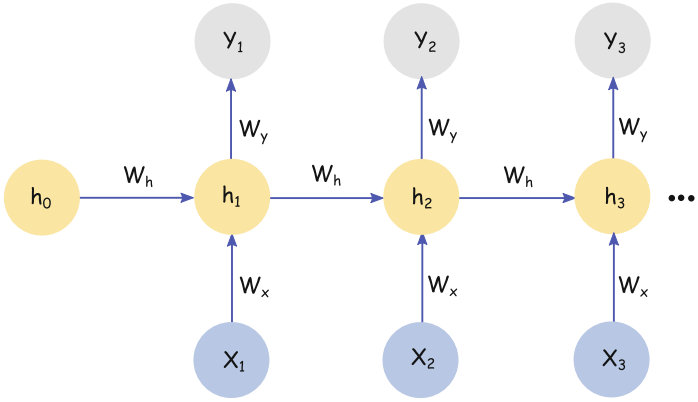


Fig. 1. An unraveled RNN where the hidden state carries important information to the next layer. Source: [12]

2.4 Long Short Term Memory (LSTM)

LSTM is a special kind of RNN capable of longer memory, thereby allowing for a more accurate model. RNNs are explicitly designed to avoid the long-term dependency problem. Instead of having one neural network, they have 4 [12].

A supervised learning approach can be taken with LSTM using word2vec. Firstly, the words in the labelled dataset are converted into vectors using word2vec and then the word sequence in sentences are input to LSTM to learn the long distance contextual dependency among words. The quality and quantity of the labelled dataset we have greatly affects the performance of the model.

2.5 Bidirectional Encoder Representations from Transformers (BERT)

BERT is a fairly new technique for NLP pre-training, released by Google in 2018 [13]. BERT is "the first bidirectional, unsupervised language representation, pre-trained using only a plain text corpus". What makes this model different is that

it takes context based on bidirectional input, i.e.: it looks at the entire sentence rather than just a word. For example, let us look at the following two sentences: He used the bow to play the violin, and, He used the bow to hunt an animal. In a uni-directional model, the model tries to assess context using only what currently exists in the sentence, so the word bow will have the same meaning in both sentences. However, in BERT the words would have different meanings based off the context of the word as well. This allows the words to have a deeper and more correct interpretation in the model. Whilst the model is trained, random words are masked and then the model tries to predict the masked word. BERT also learns relationships between sentences by detecting whether two sentences are related to each other based on context.

BERT is a very versatile model which can be fine tuned to perform sentiment analysis or even document classification. When doing this, we use BERT as a model for our supervised text data, which is then optimized to provide binary text classification based on the vectors outputted by the BERT model. BERT is able to identify context bidirectionally, so it should be able to interpret a language much better than other methods.

2.6 Convolutional Neural Network (CNN)

This is generally used for image processing, and takes an image as an input, processes it and is able to extract certain features from it or differentiate between 2 images. Although it is generally used for two dimensional image data, it can also be used with both one-dimensional and three-dimensional data. Convolution involves the multiplication of a set of weights with the input. The multiplication is performed between an array of input data and a two-dimensional array of weights, called a filter. This filter is applied on multiple points of the array resulting in a single scalar value on each application, producing another smaller matrix called a feature map. This feature map is what is used for processing [14].

2.7 Term Frequency-Inverse Document Frequency (TF-IDF)

TF-IDF is a statistical approach that identifies how important a word is to a document. This approach is based on how many times a word appears in a document, and in how many documents the same word appears. This helps separate words common in all documents and finds unique words to documents which could help classify them. This approach does not really cover the semantics of sentences, and what they mean, which makes this approach less useful for NLP [15].

Seeing that the vocabulary used in phishing and non phishing text is quite different, we decided to try TF-DIF for classification. This model might perform well in general situations for a typical phishing message, yet perhaps not as well, if phishing text is well disguised with elaborate vocabulary.

3 Methodology

We are trying to explore and compare different NLP approaches and models to correctly identify a given text as phishing or non-phishing. The process begins with the gathering of data, followed by pre-processing and initial data analysis. Then we can train, test and tune our selection of models. The models were created with the support of the following python libraries: pytorch, tensorflow, scikit-learn.

3.1 Data Collection

The collection of datasets (specially phishing text) is lengthy and difficult, so we decided to find reliable datasets available for use. We found that a majority of researches use Jose Phishing[1] as their main phishing dataset, so we decided to use it as well [16]. In this dataset, there are about ten thousand emails from 2015 to 2019 organized in different mbox files (file types which combines several emails with their headers and other information into a single file).

We also used a dataset containing fraudulent emails (Nigerian letters) found on Kaggle[2] [17] which contains 3700 emails also in the form of an mbox file. In these kind of attacks, the attacker poses as a person with wealth and power, and tries to convince the reader to send them money with the promise of great riches in return. As this type of attack is very common (specially in developing countries) and it is also a type of phishing, we decided to use it in our models.

For standard emails for comparison, we decided to go with one of the most popular datasets, the Enron corpus dataset [18]. It contains data from about 150 users, mostly senior management of Enron, organized into folders totaling around half a million emails. The emails are stored as maildir format which we converted to mbox format using Enron2mbox Script by Jimmy Lin [19]. Since a users inbox could contain phishing emails, we decided to take only their sent mail, which should contain only "legitimate" emails. There are more than 55000 emails in sent folders which should be enough for our purposes.

As we are only concerned with the text message from the datasets we have, we extracted the text mbox class using the mailbox and email module available in the Python Standard Library [20,21] and saved the text to a CSV file so it is easy to manage and work with.

3.2 Data Cleaning and Pre-processing

Data cleaning is the process of preparing data for analysis by removing any irrelevant, duplicate and incorrect data. The text we previously saved in CSV files in the previous step had a lot of HTML tags, foreign characters and several Unicode characters that was not needed for our purposes. We removed these characters as it may hinder the training process or provide inaccurate results. Figure 2 shows the different steps we took in data cleaning process.

[1] https://monkey.org/~jose/phishing/.
[2] https://www.kaggle.com/rtatman/fraudulent-email-corpus.

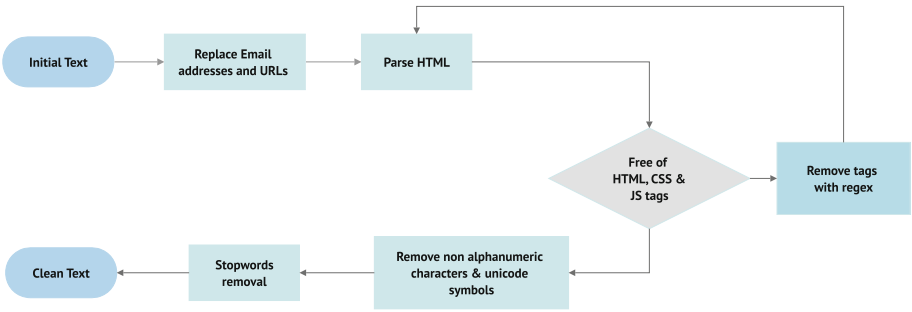


Fig. 2. Flowchart showing different steps of data cleaning process

There were several URLs and emails mixed in with the text which we did not need. We began by replacing emails with the word *email* and URLs with the word *URL*. Then we saw that the text had a lot of HTML, CSS and JavaScript tags, we used the BeautifulSoup4 HTML parser to remove those tags. We also removed other non-alphanumeric characters and also some other improperly parsed Unicode characters.

Between each step of the Data cleaning and Pre-Processing process, we made multiple analyses on the word frequency count and word clouds, to see if there is anything more that we would have to remove. Upon such analysis, we found that commonly used words; such as "the", "a", "an", "in", had highest frequency. We used the Natural Language Toolkit (NLTK) stopwords corpus to remove all those commonly used words [22]. After that we were left with cleaner text which we saved to a CSV file for latter use. We also looked into lemmatization and stemming, however we decided to not use it at this stage as it did not seem very useful. Based on the model being used, we might need to do some more pre-processing, like creating Document-Term Matrix.

3.3 Exploratory Data Analysis

Exploratory Data Analysis refers to general analysis of data often with visual methods to summarize major characteristics in a dataset. This step is helpful to identify any major trends in the data and make initial predictions before any sort of model training. This process goes hand in hand with data cleaning and pre-processing. We can use the results of Exploratory Data Analysis to further improve data cleaning and pre-processing.

Taking a look at Fig. 3, we can clearly see a distinction in the vocabulary used in phishing and non phishing texts, this leads us to believe that our models should be somewhat successful with phishing text classification. Words like bank, address, account and security are way more commonly used in phishing texts than in non phishing texts. This general difference of vocabulary will play an important role in classification of text in latter stage.

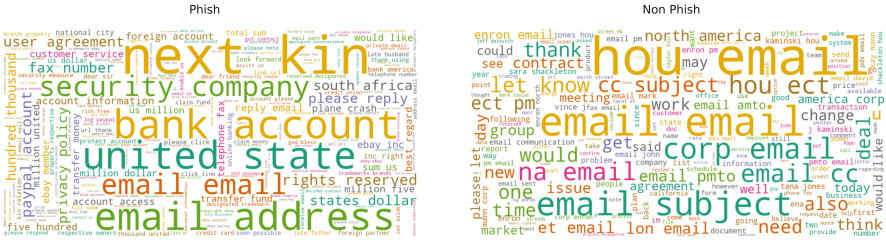


Fig. 3. Wordcloud with word frequency on phishing and non phishing text

3.4 Testing and Evaluation

To test our models performance, we feed unseen text into the model to make predictions and then check the accuracy using different metrics. We will mainly use the confusion matrix, f1-score and accuracy score to test the models performance, as these are generally industry standards.

4 Results

The same approach was followed for several algorithms and techniques, to assess the performance and quality of each one of them (Table 1).

Table 1. Results of the different tested models for classification of phishing texts

Model	F1-score	Accuracy	FP	FN
BERT	98.94	98.93	17	12
CNN + Word2Vec	98.38	98.37	16	28
LSTM	98.62	98.52	24	16
TF-IDF Classification	86.24	85.31	278	120
TF-IDF Clustering	90.03	91.26	45	211

FN/P: False Negatives/Positives.

Before implementing any of the models we were expecting BERT to be the most successful model, as it was one of the more complex models, and provided a very in depth understanding of human language, so it was no real surprise when it turned out to be the most successful model that we tested, boasting an accuracy close to 99%.

Similarly to BERT using a CNN over Word2Vec was expected to perform well, however we were not expecting it to perform so similarly to the results of BERT. This is mostly as a result of the strength of the Word2Vec model that we used, as it is a very well made model, only seconded to BERT.

Before using Doc2Vec we thought that we might see a clear definition in the vectors of texts of phishing and non-phishing, however we were surprised to see

that there was no clear split in any cluster. Although there was no clear split, a line could be drawn through the single cluster and roughly separate the data in the phishing and non-phishing texts. This can be seen in Fig. 4 and 5. This result does make sense as if it was clear to for a user to identify a phishing attack, then phishing would not be as successful as it is. This model is not very suitable for practical use, as it is the borderline phishing attacks that are difficult to detect, and are the main threats. In this model we saw that there was no real significant increase in accuracy when using more clusters, so we decided that 2 clusters were better for simplicity. We decided not to classify the texts with this approach as the test accuracy was only around 80%.

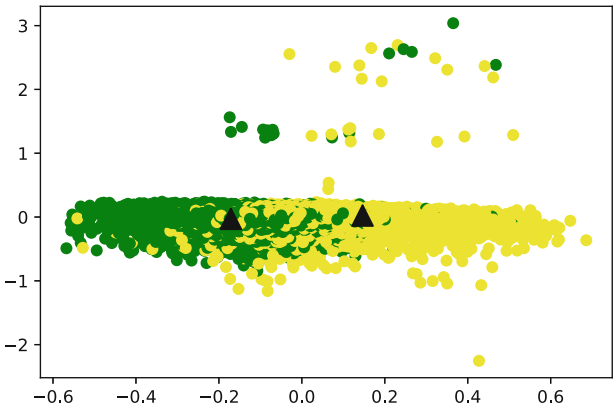


Fig. 4. A graph showing the actual Doc2Vec vectors for each document, where green dots represent phishing texts, and yellow represents non-phishing texts. As can be seen, a vertical split can be made to split majority of the texts from each other (Color figure online)

When using **TF-IDF** with classification we were not expecting very high results, we just included this model as a purely mathematical approach to the problem domain, and we were pleasantly surprised to see that the model performed so well. This could be as a result of the difference in the texts of the email, i.e. certain words are generally used a lot in phishing emails and hardly in normal emails.

We also decided to try k-means clustering with **TF-IDF** and our best results were with approximately 9 clusters. The clusters can be seen in Figs. 6 and 7 we would assign a text to a cluster and would assign the text as either phishing or non-phishing depending on what the majority of texts were in that cluster. What was very interesting with this approach, is that a lot of the clusters were very one sided, indicating that there are certain characteristics that can separate some phishing texts and vice versa, however due to the large amount of False negatives this model is still not very optimal to use.

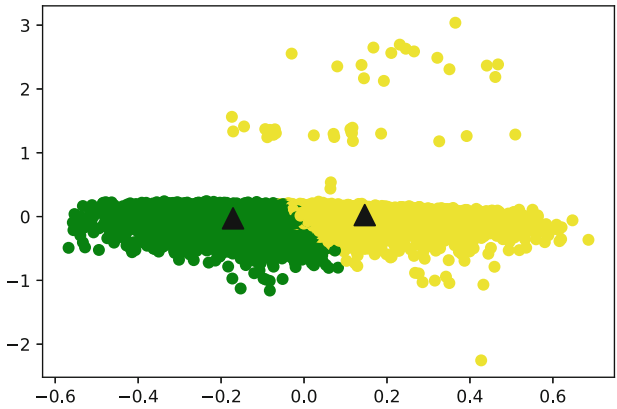


Fig. 5. A graph showing unclassified k-means clustering on DOC2VEC values. As can be seen the split made here looks similar to the graph on the left, thereby indicating that we could cluster text with some accuracy, roughly 80%

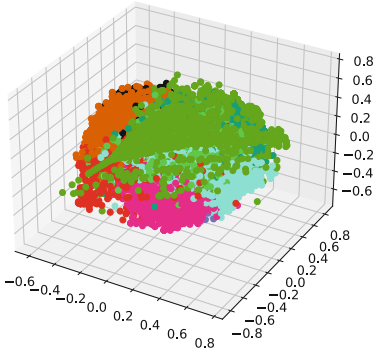


Fig. 6. A 3d plot showing the different texts in clusters, each color relating to a different cluster.

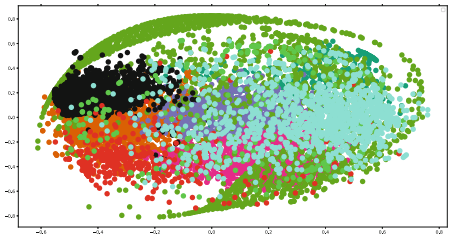


Fig. 7. A 2d plot showing the different texts in clusters, each color relating to a different cluster.

5 Conclusion

Our main concern with our study is that we are not sure that the results are accurate reflections of real world data, however there is nothing that we can really do to find better datasets. We believe there is a clear difference between the texts that were phishing and the texts that were non-phishing, making it easier for the models to classify the texts.

There is a lot more fine tuning which could have been done on the models to improve the results even more, but we believed that as a proof of concept it was not needed to improve the models, especially since a lot of them had better results than we expected.

As seen it is clear that BERT has the best performance out of all the models tested, furthermore it has the lowest False Negatives, which we believe to be an important factor in our use case. It is important for our model to rather over classify texts as phishing than to under-classify them.

With the results of this project, we believe that NLP can help detect phishing if used in conjunction with other factors, this could include any URLs in the text or information regarding how the text was received. We are already working on using some of these features to create an application that can sort your inbox and move possible phishing emails to a separate folder. It could be used as a client side tool to fight phishing in future.

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